

Course Code	Course Title	L	T	P	C
BACSE106	Operating Systems	3	0	2	4
Pre-requisite	NIL	Syllabus Version			
		1.0			
Course Objectives					
1. To introduce the fundamental concepts, design issues, and core functionalities of operating systems. 2. To build skills in managing processes, memory, files, devices, and ensuring protection and security, using algorithms and practical simulations. 3. To introduce modern developments like virtualization, distributed operating systems, and cloud-based systems.					
Course Outcomes					
1. Understand the structure, functionalities, and foundational concepts of operating systems. 2. Apply process scheduling algorithms and analyze thread models and deadlock-handling methods. 3. Implement synchronization mechanisms and solve concurrency problems in multi-process systems. 4. Evaluate memory management and file system strategies used in operating systems. 5. Analyze device management, protection mechanisms, and virtualization techniques in operating systems.					
Module:1	Operating Systems and Structure	6 hours			
Operating system basics - History of operating systems - OS structure: Monolithic, Layered, Microkernel, Modular - OS services - System calls and types - Interrupts - OS design issues - Building and booting an OS.					
Module:2	Process, Scheduling, Threads and Deadlocks	10 hours			
Process concepts and states - Process Scheduling - Operation on Processes: Creation, Termination and Cooperation - Threads and multithreading models - Scheduling criteria - CPU scheduling: Non-preemptive (FCFS, SJF, Priority), Preemptive (Round Robin, SRTF, Preemptive Priority), Multi-level Queue, Multi-level Feedback Queue - Deadlock characterization - Resource Allocation Graphs (RAG) - Deadlock handling methods.					
Module:3	Process Synchronization	9 hours			
IPC: Shared memory, message passing - Race condition - Critical section problem - Peterson's solution - Bakery Algorithm - Hardware synchronization: Test-and-Set, Swap - Mutex locks - Semaphores - Classical synchronization problems: Bounded buffer, Readers-Writers, Dining Philosophers - Monitors.					
Module:4	Memory and File Management	10 hours			

Address binding - Logical and physical address space - Contiguous memory allocation - Fragmentation - Placement Algorithms: First Fit, Best Fit, Worst Fit - Paging: Structure with TLB - Segmentation - Virtual memory: Demand paging - Page replacement: FIFO, Optimal, LRU - Thrashing and Working Set - File system concepts - File access methods - Directory structures and implementation - File allocation methods.		
Module:5	Managing Devices, Security and Protection and Virtualization	8 hours
I/O device management basics: RAID structure, Disk structure, Disk Scheduling Algorithms (FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK), Program and network threats - Cryptography as a security tool - Domains of protection - Access matrix - Capability based systems - Need for virtualization - Virtual machines and architectures - Hypervisors - Virtualization Technologies: Para Virtualization, Full Virtualization - OS-Level Virtualization: Containers, Docker Basics, Containers vs Virtual Machines.		
Module:6	Contemporary Issues	2 hours
Total Lecture Hours:		45 hours
Text Book(s)		
1. Abraham Silberschatz, Peter B. Galvin, Greg Gagne, " Operating System Concepts ", Wiley, 10 th Edition, 2018		
Reference Books		
1. Remzi H. Arpaci-Dusseau, Andrea C. Arpaci-Dusseau, " Operating Systems: Three Easy Pieces ", Arpaci Dusseau Books, 1 st Edition, 2023 2. Andrew S. Tanenbaum, Herbert Bos, " Modern Operating Systems ", Pearson Education, 4 th Edition, 2015 3. William Stallings, " Operating Systems: Internals and Design Principles ", Pearson, 9 th Edition, 2018		
Indicative Experiments		
1. Study of basic Linux commands and shell scripting		2 hours
2. Implementation of Bootloader		2 hours
3. Process creation using fork(), Orphan and Zombie processes		2 hours
4. Thread Creation		2 hours
5. CPU scheduling algorithms (FCFS, SJF, Priority)		2 hours
6. CPU scheduling algorithms (Round Robin, SRTF, Pre-emptive Priority)		2 hours
7. Banker's Algorithm for Deadlock Avoidance (Safe state checking and additional resource request granting)		2 hours

8. Shared Memory Creation and sharing contents between processes	2 hours
9. Implementation of Producer-Consumer/Readers Writers using Semaphore	2 hours
10. Implementation of Dining Philosophers using Semaphore	2 hours
11. Memory Allocation Strategies: First Fit, Best Fit, Worst Fit	2 hours
12. Page Replacement Algorithms: FIFO, LRU, Optimal	2 hours
13. Disk Scheduling Algorithms: FCFS, SSTF, SCAN	2 hours
14. Disk Scheduling Algorithms: C-SCAN, LOOK, C-LOOK	2 hours
15. File system operations using system calls	2 hours
Total Laboratory Hours:	30 hours
Mode of Evaluation :Continuous Assessment Test, Digital Assignment, Quiz, Final Assessment Test, Lab Continuous Assessment, Lab Final Assessment	
Recommended by Board of Studies :	21-05-2025
Approved by Academic Council : No. 78	12-06-2025